

Hierarchical Bayesian Model Specification

Response Variable:

The dependent variable in the Bayesian regression model is the monthly dividend-adjusted log returns for individual S&P 500 stocks:

$$r_{i,t} = \alpha_i + X'_{i,t}\beta + \varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim N(0, \sigma_i^2)$$

Where:

$r_{i,t}$: Return for stock i at time t

$X_{i,t}$: Vector of predictors (factors) for stock i at time t

β : Vector of regression coefficients

$\varepsilon_{i,t}$: Normally distributed error term with variance σ_i^2

Prior Distributions (Hierarchical Structure & Spike-and-Slab):

1. Hierarchical Priors by Economic Themes:

For each coefficient $\beta_{j,k}$ (factor j within theme k):

$$\beta_{j,k} \sim N(\mu_k, \tau_k^2)$$

Here, each factor within a theme shares a common distribution, allowing information pooling and thematic coherence.

- Hyperpriors for Theme Means:
 - $\mu_k \sim N(0, \sigma_\mu^2)$
- Hyperpriors for Theme Variances:
 - $\tau_k \sim HalfCauchy(scale_\tau)$

Spike-and-Slab Prior on Individual Factors:

To facilitate automatic feature selection at the individual factor level, a spike-and-slab prior is applied:

$$\beta_{j,k} \sim \gamma_{j,k} \cdot N(0, s_{slab}^2) + (1 - \gamma_{j,k}) \cdot N(0, s_{spike}^2)$$

- S_{spike} : very small variance, tightly concentrated around zero

- S2slab: Larger variance, allowing significant deviation from zero.
- Inclusion indicator $\gamma_{j,k} \sim \text{Bernoulli}(\pi_{j,k})$: determining factor relevance. A high gamma means the factor as been 'selected'. Significant contributor to return forecasting.

Novel Approach: Informing Prior Inclusion Probabilities (gamma) Using Jensen et al. (2023):

The innovative part of this approach is setting prior inclusion probabilities informed by the economic significance identified by Jensen et al. (2023):

- If Jensen et al. found theme to be economically highly significant, factors within this theme are given a higher prior probability of inclusion:
 $\pi_{j,k} \sim \text{Beta}(2,1)$ (or similar, indicating higher prior inclusion probability)
- Conversely, for themes Jensen et al. deemed less relevant, a lower prior probability of inclusion is assigned:
 $\pi_{j,k} \sim \text{Beta}(1,2)$ (or similar, indicating lower prior inclusion probability)

Variance Prior:

Error variances σ_i^2 can initially be assumed common for simplicity or allowed to vary by stock with the following prior distribution:

$$\sigma_i^2 \sim \text{InverseGamma}(a_\sigma, b_\sigma)$$

Posterior Distribution Estimation:

Using Bayes' theorem, the posterior distribution combines the data and the prior information to provide the updated distribution of the parameters:

$$p(\beta, \gamma, \mu, \tau, \sigma^2 | r, X) \propto p(r | X, \beta, \sigma^2) \cdot p(\beta | \gamma, \mu, \tau) \cdot p(\gamma | \pi) \cdot p(\mu) \cdot p(\tau) \cdot p(\sigma^2)$$

Python Implementation:

The spike-and-slab hierarchical Bayesian model will be implemented using the PyMC Python library, a powerful tool for Bayesian statistical modeling and Markov Chain Monte Carlo

(MCMC) sampling, offering straightforward methods to specify complex priors and hierarchical structures clearly.

Purpose:

This detailed mathematical formulation is aimed at obtaining precise feedback regarding the Bayesian hierarchical structure, the spike-and-slab priors tailored with economic insights from Jensen et al. (2023), and the overall suitability of these modeling choices for my forecasting objectives.